الأسئلة المقالية — 11 عام · مذاكرة الهيكل الاختبار النهائي للفصل الاول

الاسم: _____ المدرسة: _____ التاريخ: _____ (32 سؤالاً)

LO21 خصائص المضلعات المتشابهة · Properties of similar polygons

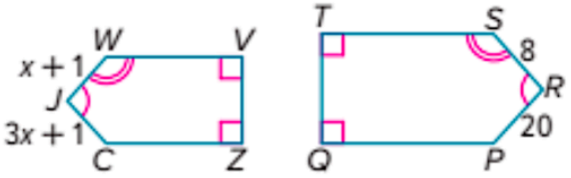
21.1 LO21 · 21a

Complete every part of the question shown in the image. Write your working in the space below.

الأستاذ كامل بشارة 0502837825

21a The pair of polygon is similar. Find the value of x

اذا كان الشكلان متشابهان. اوجد قيمة x



a. Complete all parts of the question.

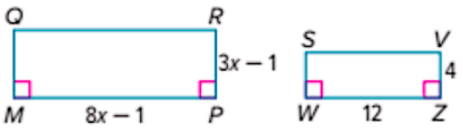
الأستاذ كامل بشارة 0502837825

الأستاذ كامل بشارة 0502837825

21.2 LO21 · 21b

Complete every part of the question shown in the image. Write your working in the space below.

21b The pair of polygon is similar. Find the value of x	اذا كان الشكلان متشابهان. اوجد قيمة x

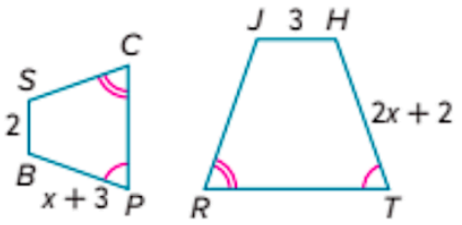


a. Complete all parts of the question.

21.3 LO21 · 21c

Complete every part of the question shown in the image. Write your working in the space below.

21c The pair of polygon is similar. Find the value of x	اذا كان الشكلان متشابهان. اوجد قيمة x

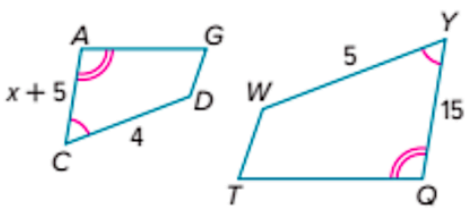


a. Complete all parts of the question.

21.4 LO21 · 21d

Complete every part of the question shown in the image. Write your working in the space below.

21d The pair of polygon is similar. Find the value of x	اذا كان الشكلان متشابهان. اوجد قيمة x



a. Complete all parts of the question.

21.5 LO21 ·

Rectangles ABCD and EFGH are similar in that order. (a) Find x. (b) Find the scale factor from ABCD to EFGH. Show proportions.

$$AB = 6,\ BC = x + 1,\ EF = 15,\ FG = 20$$

a. Proportion and x

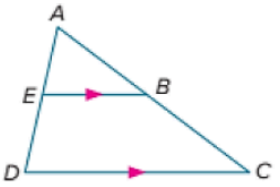
b. Scale factor

22.1 LO22 · 22a

Complete every part of the question shown in the image. Write your working in the space below.

22a If $AB = 6, BC = 4$ $and\ AE = 9$

$find\ ED$



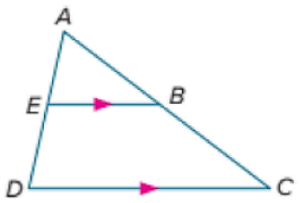
a. Complete all parts of the question.

22.2 LO22 · 22b

Complete every part of the question shown in the image. Write your working in the space below.

22b If $AB = 12$, $AC = 16$ and $ED = 5$

$find\ AE$



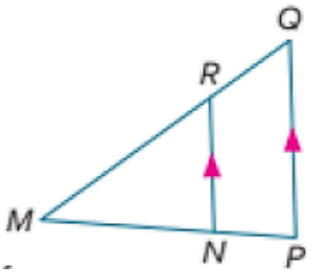
a. Complete all parts of the question.

22.3 LO22 · 22c

Complete every part of the question shown in the image. Write your working in the space below.

22c Determine whether $\overline{NR} \parallel \overline{PQ}$.

$$PM = 18, PN = 6, QM = 24, and\ RM = 16$$



a. Complete all parts of the question.

22.4 LO22 · 22d

M,N,P are collinear in that order, and M,R,Q are collinear in that order. Determine whether NR is parallel to PQ. Justify using ratios.

$$QM = 31,\ RM = 21,\ PM = 4PN$$

a. Complete all parts of the question.

22.5 LO22 ·

In triangle ABC, D lies on AB and E lies on AC. (a) Determine whether DE is parallel to BC. (b) Justify your answer using the converse of the Triangle Proportionality Theorem.

$$AD = 6,\ DB = 9,\ AE = 8,\ EC = 12$$

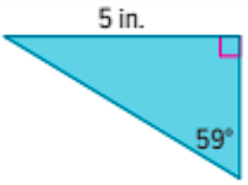
a. Compare ratios

b. Conclusion with reason

23.1 LO23 · 23a

Complete every part of the question shown in the image. Write your working in the space below.

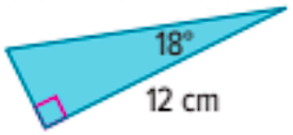
23a Find the perimeter and area to the nearest hundredth	اوجد محيط ومساحة الشكل التالي لأقرب جزء من مئة



a. Complete all parts of the question.

23.2 LO23 · 23b

Complete every part of the question shown in the image. Write your working in the space below.

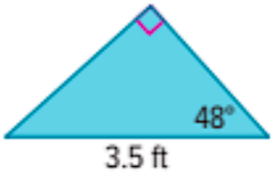
23b Find the perimeter and area to the nearest hundredth	اوجد محيط ومساحة الشكل التالي لأقرب جزء من مئة
	

a. Complete all parts of the question.

23.3 LO23 · 23c

Complete every part of the question shown in the image. Write your working in the space below.

23c Find the perimeter and area to the nearest hundredth	اوجد محيط ومساحة الشكل التالي لأقرب جزء من مئة



a. Complete all parts of the question.

23.4 LO23 ·

A right triangle has hypotenuse 12 cm and an acute angle of 35°. (a) Find both legs. (b) Find its perimeter and area. Round final results to two decimal places.

a. Two legs

b. Perimeter

c. Area

23.5 LO23 ·

A right triangle has hypotenuse 14 cm and an acute angle of 40°. (a) Find both legs. (b) Find its perimeter and area. Round final results to two decimal places.

a. Two legs

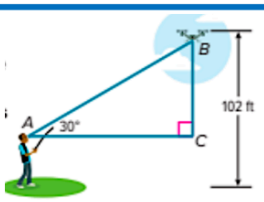
b. Perimeter

c. Area

24.1 LO24 · 24a

Complete every part of the question shown in the image. Write your working in the space below.

24a Rakeem is flying his drone at the park. He spots the drone at an angle of elevation that he estimates to be 30°. The remote control tells Rakeem that his drone is 102 feet above the ground. If Rakeem is 6 feet tall, how far is he from the drone to the nearest foot?	يُطيّر راكيم طائرته المسيّرة في الحديقة. يرى الطائرة بزاوية ارتفاع يقدّرها بـ30°. يُخبر جهاز التحكم عن بُعد راكيم أن طائرته ترتفع 102 قدمًا عن سطح الأرض. إذا كان طول راكيم 6 أقدام، فما المسافة بينه وبين الطائرة المسيّرة، مقربةً إلى أقرب قدم؟

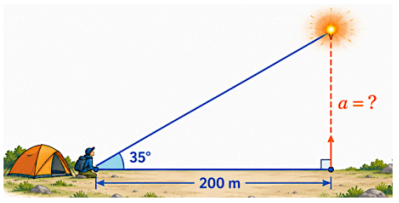


a. Complete all parts of the question.

24.2 LO24 · 24b

Complete every part of the question shown in the image. Write your working in the space below.

24b A flare is shot vertically into the air approximately 200 meters from base camp. The angle of elevation to the maximum height of the flare is 35°. The group at base camp needs to know the altitude of the flare.What is the maximum height of the flare to the nearest meter?	تُطلَق شعلة إشارة رأسيًا في الهواء من نقطة تبعد نحو 200 متر عن المخيم الأساسي. تبلغ زاوية الارتفاع إلى أعلى نقطة تصل إليها الشعلة 35°. تحتاج المجموعة الموجودة في المخيم الأساسي إلى معرفة ارتفاع الشعلة. ما أقصى ارتفاع تصل إليه الشعلة، مقربًا إلى أقرب متر؟



a. Complete all parts of the question.

24.3 LO24 ·

A student observes the top of a vertical tower at an elevation angle of 35°. The student is 35 m from its base on level ground, and their eyes are 1.6 m above the ground. (a) Write an equation. (b) Find the tower height to the nearest tenth of a metre.

a. Equation

b. Height

24.4 LO24 ·

A student observes the top of a vertical tower at an elevation angle of 40°. The student is 40 m from its base on level ground, and their eyes are 1.6 m above the ground. (a) Write an equation. (b) Find the tower height to the nearest tenth of a metre.

a. Equation

b. Height

24.5 LO24 ·

A student observes the top of a vertical tower at an elevation angle of 45°. The student is 45 m from its base on level ground, and their eyes are 1.6 m above the ground. (a) Write an equation. (b) Find the tower height to the nearest tenth of a metre.

a. Equation

b. Height

25.1 LO25 · 25a

Complete every part of the question shown in the image. Write your working in the space below.

25a Sunita and Derek work for a company that produces microchips. Part of their job is to estimate the number of defective chips given the total number produced. The table shows the estimated contents of 2 boxes of microchips.Derek randomly selects one chip from Box B, does not put it back, and then randomly selects another chip from Box B. What is the probability that both chips are defective? Explain.

تعمل سونيتا وديريك في شركة تُنتج رقائق إلكترونية. ويتمثّل جزء من عملهما في تقدير عدد الرقائق المعيبة بناءً على العدد الإجمالي المُنتَج. يوضّح الجدول المحتويات التقديرية لصندوقين من الرقائق الإلكترونية. يسحب ديريك عشوائيًا رقاقة واحدة من الصندوقB ، ولا يعيدها، ثم يسحب عشوائيًا رقاقة أخرى من الصندوق نفسه. ما احتمال أن تكون الرقيقتان معيبتين؟ وضّح إجابتك.

	Number of Chips	Defective
Box A	100	4%
Box B	150	2%

a. Complete all parts of the question.

25.2 LO25 · 25b

Box A contains 100 microchips, 4% defective. Box B contains 150 microchips, 2% defective. Sunita selects a chip from Box A, replaces it, and selects another from Box A. A claim states that the probability both chips are defective is 1/125. Is that claim correct? Calculate the probability and explain.

a. Complete all parts of the question.

25.3 LO25 · 19

Sunita and Derek work for a microchip company. They estimate defective chips using the contents of two boxes shown in the table. Sunita randomly selects one chip from Box A and then randomly selects another chip from Box A. The probability that both chips are defective is1/625. Did Sunita replace the first chip before selecting the second one? Explain.

Box	Number of chips	Defective
A	100	4%
B	150	2%

a. Compare with replacement

b. Compare without replacement

c. Conclusion

25.4 LO25 · 20

TRAVEL. A travel agency surveys families to determine whether they drive (D) or fly (F) to their vacation destinations. The results show P(D)=0.6 and P(D and F)=0.2. The probability that a family did not vacation is0.1. (a) What is the probability that a family reached its vacation destination by flying? (b) What is the probability that a family that drives will also fly?



a. (a) Probability of flying

b. (b) Conditional probability of flying given driving

25.5 LO25 ·

A bag contains 4 red counters and 6 blue counters. Two are drawn without replacement. (a) Explain whether the draws are independent. (b) Find P(both red). (c) Find P(both red) if the first counter is replaced before the second draw.

a. Dependence

b. Without replacement

c. With replacement

25.6 LO25 ·

A bag contains 5 red counters and 7 blue counters. Two are drawn without replacement. (a) Explain whether the draws are independent. (b) Find P(both red). (c) Find P(both red) if the first counter is replaced before the second draw.

a. Dependence

b. Without replacement

c. With replacement

25.7 LO25 ·

A bag contains 6 red counters and 8 blue counters. Two are drawn without replacement. (a) Explain whether the draws are independent. (b) Find P(both red). (c) Find P(both red) if the first counter is replaced before the second draw.

a. Dependence

b. Without replacement

c. With replacement

26.1 LO26 · 25c

Complete every part of the question shown in the image. Write your working in the space below.

25c A travel agency conducts a survey to determine whether people drive (D) or fly (F) to their vacation destinations. The results indicated that P(D)=0.6, P(D∩ F)=0.2 , and the probability that a family did not vacation is 0.1

تُجري وكالة سفر استطلاعًا لتحديد ما إذا كان الناس يسافرون بالسيارة (D) أو بالطائرة (F) إلى وجهات عطلاتهم. أظهرت النتائج أن

P(D)=0.6، P(D∩ F)=0.2، وأن احتمال عدم ذهاب الأسرة في عطلة هو 0.1

a. What is the probability that a family reached their vacation destination by flying?

ما احتمال أن تكون الأسرة قد وصلت إلى وجهة عطلتها بالطائرة؟



b. What is the probability that a family that drives will also fly?

ما احتمال أن تسافر الأسرة بالطائرة أيضًا، علمًا بأنها تسافر بالسيارة؟

a. Complete all parts of the question.

26.2 LO26 · 26a

Complete every part of the question shown in the image. Write your working in the space below.

26a A game piece is selected at random from the plate at the right. What is the probability that the game piece is round or orange?	تُختار قطعة لعب عشوائيًا من الطبق الموجود على اليمين. ما احتمال أن تكون قطعة اللعب دائرية أو برتقالية اللون؟
	

a. Complete all parts of the question.

26.3 LO26 · 26b

Complete every part of the question shown in the image. Write your working in the space below.

26b in a group of graduate students, 4 out of the 5 females are international students, and 2 out of the 3 men are international students. What is the probability of selecting a graduate student from this group that is a male or an international student?	في مجموعة من طلاب الدراسات العليا، هناك 4 طالبات دوليات من أصل 5 طالبات، وطالبان دوليان من أصل 3 طلاب ذكور. ما احتمال اختيار طالب من هذه المجموعة يكون ذكرًا أو طالبًا دوليًا؟

a. Complete all parts of the question.

26.4 LO26 ·

In a group of 35 students, 13 study art, 16 study music, and 6 study both. (a) Find the probability a random student studies art or music. (b) Find the probability of neither. (c) Explain why the events are not mutually exclusive.

a. Union

b. Complement

c. Explain overlap

26.5 LO26 ·

In a group of 40 students, 14 study art, 17 study music, and 7 study both. (a) Find the probability a random student studies art or music. (b) Find the probability of neither. (c) Explain why the events are not mutually exclusive.

a. Union

b. Complement

c. Explain overlap